MIRAGE: MRI-Informed Representation for Alzheimer's Graph Embeddings — A Semi-Supervised Topological Neuroimaging Framework

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***Abstract*—The classification of Alzheimer's Disease (AD) from magnetic resonance imaging (MRI) has traditionally relied on high-dimensional convolutional neural networks or flat tabular machine learning baselines, both of which often lack structural interpretability. In this paper, we propose MIRAGE (MRI-Informed Representation for Alzheimer's Graph Embeddings), a novel two-stage semi-supervised framework that combines self-supervised image representation learning with graph-based tissue topology modeling. Using 161 labeled and 135 unlabeled subjects from the OASIS-1 dataset, we first pretrain a Variational Autoencoder (VAE) on 2.5D multi-planar MRI slices across all 296 subjects, then use the learned encoder as a node feature extractor for a Graph Attention Network (GAT). Our baseline GAT with hand-crafted volumetric features achieves a ROC-AUC of 0.8360, significantly outperforming Logistic Regression (0.7933) under strict 5-fold cross-validation. MIRAGE's image-derived embeddings achieve a combined ROC-AUC of 0.831, statistically equivalent to the hand-crafted baseline, demonstrating that self-supervised pretraining on neuroimaging data can match expert-designed features even at small dataset scales. Through attention weight analysis, the GAT independently localizes disease pathology by assigning maximum predictive importance to CSF expansion (mean attention weight 0.3965), mathematically mirroring the clinical phenomenon of hydrocephalus ex vacuo. These findings advance both the accuracy and interpretability of automated AD diagnosis while providing empirical evidence on the limits of learned representations in small-cohort neuroimaging studies.**

***Index Terms*—Graph Attention Networks, Alzheimer's Disease, Variational Autoencoder, Semi-Supervised Learning, Neuroimaging, Deep Learning, Interpretability**

I. Introduction

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder characterized by gradual cognitive decline and severe brain atrophy. Early and accurate diagnosis relies heavily on clinical assessments paired with structural Magnetic Resonance Imaging (MRI). As the global incidence of dementia rises, the demand for scalable, automated, and interpretable computer-aided diagnostic systems has become paramount.

Historically, machine learning approaches for AD classification have diverged into two methodologies. The first utilizes tabular clinical and demographic data fed into traditional classifiers such as SVMs or Random Forests. While computationally lightweight, these models discard spatial and structural relationships within the brain. The second employs 3D Convolutional Neural Networks (CNNs) on raw MRI voxels. While often accurate, 3D CNNs are computationally expensive, prone to overfitting on small medical datasets, and function largely as black boxes [1].

Graph Neural Networks (GNNs), and specifically Graph Attention Networks (GATs), offer a compelling geometric alternative [2]. By representing the brain as a topological graph, where regions of interest operate as nodes and their spatial relationships as edges, we can preserve structural geometry while maintaining parameter efficiency.

In this study, we propose MIRAGE, which extends the baseline GAT framework with a semi-supervised pretraining strategy. We hypothesize that (1) translating MRI segmentations into topological micro-graphs yields superior classification metrics compared to flat tabular models, and (2) self-supervised image representation learning can replace hand-crafted volumetric features without loss of classification performance, even at the challenging scale of $N = 161$ labeled subjects.

II. Related Work

A. Classical Machine Learning in Neuroimaging

Early approaches to AD classification relied on manual extraction of volumetric and morphometric features from structural MRI. Researchers utilized tools such as FreeSurfer and FSL to extract cortical thickness, surface area, and regional volumes [4]. SVMs were particularly prevalent due to their efficacy in high-dimensional, low-sample-size regimes [5]. While interpretable and computationally efficient, these models treat brain regions as independent variables, failing to capture the interconnected topological degradation characteristic of neurodegeneration.

B. Convolutional Neural Networks

With the advent of deep learning, 3D CNNs became the de facto standard for MRI analysis, learning hierarchical spatial representations directly from raw voxel intensities

[6]. However, their application is hindered by computational bottlenecks and severe overfitting when applied to datasets with fewer than a thousand subjects. Furthermore, while techniques like Grad-CAM [7] offer post-hoc explanations, they provide limited insight into structural connectivity driving the pathology.

C. Graph Neural Networks in Medical Imaging

GNNs have emerged as a powerful paradigm bridging geometric awareness with parameter efficiency. Parisot et al. pioneered GCNs for disease prediction by modeling subject populations as graphs [8]. Veličković et al. introduced GATs with learnable attention coefficients between connected nodes [2]. While GATs have been explored in functional MRI connectomics, their application to structural MRI volumetrics with image-derived node features remains sparse. This study directly addresses this gap.

D. Semi-Supervised Learning in Medical Imaging

The scarcity of labeled medical data has motivated extensive research into semi-supervised learning. VAEs [10] have been successfully applied as pretraining mechanisms, learning compact latent representations from unlabeled data before fine-tuning on labeled subsets. Our work extends this paradigm to neuroimaging by using VAE-learned tissue embeddings as node features in a graph classifier, leveraging all available subjects regardless of label availability.

III. Methodology

A. Data Acquisition and Cohort

Data was obtained from the Open Access Series of Imaging Studies (OASIS-1) cross-sectional dataset. Subjects were filtered based on the availability of FSL segmentation masks registered to the Talairach 1988 standard stereotaxic space, mapping all images to a uniform 1mm$\times$1mm$\times$1mm isotropic grid.

The final labeled cohort consisted of $N = 161$ subjects. Ground truth labels were derived from the Clinical Dementia Rating (CDR), with $CDR = 0.0$ categorized as Healthy Controls ($n = 93$) and $CDR > 0.0$ categorized as AD patients ($n = 68$). An additional 135 subjects with valid MRI volumes but missing CDR scores were retained for unsupervised pretraining, yielding a total of 296 subjects available for representation learning.

Table I details the demographic and clinical characteristics of the labeled cohort.

Notably, 72.1% of AD patients present with CDR=0.5 (very mild dementia), representing the most clinically challenging classification boundary where subtle atrophy patterns overlap significantly with age-related brain volume loss in healthy elderly subjects.

TABLE I
Cohort Demographics and Clinical Characteristics

Characteristic	Healthy Controls (n=93)	AD Patients (n=68)
Age (Years)	72.8 $\pm$ 8.6	76.2 $\pm$ 7.1
Gender (M/F)	25 / 68	29 / 39
Education (Years)	3.2 $\pm$ 1.3	2.7 $\pm$ 1.4
MMSE Score	28.9 $\pm$ 1.2	24.3 $\pm$ 4.1
eTIV (cm^3)	1445.6 $\pm$ 156.2	1478.3 $\pm$ 164.7
nWBV	0.758 $\pm$ 0.035	0.713 $\pm$ 0.038
Clinical Dementia Rating (AD Patients)		
CDR = 0.5 (Very Mild)	—	49 (72.1%)
CDR = 1.0 (Mild)	—	18 (26.5%)
CDR = 2.0 (Moderate)	—	1 (1.4%)

B. Baseline Tabular Modeling

To establish a statistical benchmark, a feature matrix was constructed using tabular metadata: Age, Education, SES, eTIV, nWBV, and ASF. Missing SES values were handled via median imputation, and continuous variables were normalized using a standard scaler. We evaluated Logistic Regression, SVM (RBF kernel), and Random Forest using Stratified 5-Fold Cross-Validation.

C. Stage 1: 2.5D Slice Extraction

To bridge raw MRI volumes and the graph framework, we developed a compact 2.5D multi-planar representation. For each of the 296 subjects, we loaded the skull-stripped, atlas-registered MRI volume (`_t88_masked_gfc.img`, shape $176 \times 208 \times 176$, 1mm isotropic) and the corresponding FSL segmentation mask.

For each tissue type $t \in \{\text{CSF}, \text{GM}, \text{WM}\}$:

1) A binary tissue mask was constructed from the segmentation labels (1=CSF, 2=GM, 3=WM)
2) The MRI volume was masked element-wise, preserving only voxel intensities belonging to tissue t
3) The anatomical center of mass $\mathbf{c}_t = (c_x, c_y, c_z)$ was computed via `scipy.ndimage.center_of_mass`
4) Three orthogonal 2D slices were extracted at $\mathbf{c}_t$: axial, coronal, and sagittal planes
5) Each slice was resized to 64×64 pixels via bicubic interpolation
6) The three slices were stacked as a 3-channel image of shape $(3, 64, 64)$

Per-subject intensity normalization was applied prior to slicing: $\hat{x} = (x - \mu_{\text{brain}})/\sigma_{\text{brain}}$, where statistics were computed over non-zero (brain tissue) voxels only. The final dataset comprised $296 \times 3 = 888$ samples of shape $(3, 64, 64)$, stored as a compressed 18.6 MB `.npz` archive.

D. Stage 2: Variational Autoencoder Pretraining

A Variational Autoencoder (VAE) was pretrained on all 888 tissue samples using a self-supervised reconstruction objective. No CDR labels were used during pretraining, enabling utilization of all 296 subjects including the 135 without diagnostic labels.

The BrainVAE architecture consists of an encoder compressing input $(3, 64, 64)$ images to a 64-dimensional latent

distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2\mathbf{I})$, and a symmetric decoder reconstructing the original image. The encoder comprises three convolutional blocks (channels: $3 \to 32 \to 64 \to 128$, stride=2), a fully connected layer mapping to 256 hidden units, and two parallel linear heads producing $\boldsymbol{\mu} \in \mathbb{R}^{64}$ and $\log \boldsymbol{\sigma}^2 \in \mathbb{R}^{64}$.

The VAE loss combines reconstruction fidelity with latent space regularization:

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}[\|\mathbf{x} - \hat{\mathbf{x}}\|^2] + \beta \cdot D_{\text{KL}}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) \| \mathcal{N}(\mathbf{0}, \mathbf{I})) \quad (1)$$

where $\beta = 0.5$ balances reconstruction quality against latent space regularity. Training used Adam (lr $= 10^{-3}$, weight decay 10^{-4}) for 100 epochs with ReduceLROnPlateau scheduling, achieving best validation loss of 0.0509 at epoch 87.

E. Stage 3: Supervised Fine-Tuning

Following unsupervised pretraining, we performed supervised fine-tuning of the VAE encoder on the 161 labeled subjects. To prevent catastrophic forgetting of low-level visual features, all convolutional layers were frozen. Only the final fully-connected layers (`fc_enc`, `fc_mu`, `fc_logvar`, totaling 2,130,304 parameters) were updated.

A lightweight classification head — Linear(69, 32) → ELU → Dropout(0.5) → Linear(32, 1) — was trained jointly, receiving the mean-pooled tissue embedding concatenated with 5 normalized clinical features (Age, Education, eTIV, nWBV, Age×nWBV). Differential learning rates were applied: $\text{lr}_{\text{encoder}} = 10^{-4}$, $\text{lr}_{\text{head}} = 10^{-3}$. Early stopping monitored validation AUC with patience of 10 epochs, achieving best validation AUC of 0.8045 at epoch 28.

F. Topological Graph Construction

For the geometric deep learning pipeline, each of the 161 labeled subjects was modeled as a fully connected directed micro-graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$:

- **Nodes ($\mathcal{V}$):** Three anatomical nodes: CSF, Gray Matter, White Matter
- **Edges ($\mathcal{E}$):** Fully connected topology (excluding self-loops), yielding 6 directed edges
- **Node Features (X):** Two variants were evaluated:
  - *Baseline GAT*: 5-dimensional hand-crafted vector [tissue volume, Age, Educ, eTIV, nWBV]
  - *MIRAGE*: 64-dimensional VAE embedding from the fine-tuned encoder
- **Graph-level Features:** 5 normalized clinical covariates [Age, Educ, eTIV, nWBV, Age×nWBV] injected after global pooling, bypassing the attention mechanism entirely

G. MIRAGE Architecture

The MIRAGE model extends the baseline GAT with image-derived node features and a dual-stream classifier. The graph stream processes VAE embeddings through two GAT layers, while the clinical stream bypasses the graph to inject demographic context at the classification stage.

Formally, given node feature matrix $\mathbf{X} \in \mathbb{R}^{3\times 64}$, the attention coefficient for edge (i, j) is:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_j]))}{\sum_{k\in\mathcal{N}(i)} \exp(\text{LeakyReLU}(\mathbf{a}^T[\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_k]))} \quad (2)$$

Layer 1: GATConv($64 \to 32$, heads=4, concat=True) → ELU → Dropout(0.3), output shape $(3, 128)$

Layer 2: GATConv($128 \to 32$, heads=1) → Global Mean Pooling → $(32,)$

Classifier: Concatenate clinical features → $(37,)$ → Linear(37, 16) → ELU → Dropout(0.2) → Linear(16, 1)

The full pipeline is formalized in Algorithm 1.

Algorithm 1 MIRAGE Forward Pass

Require: Tissue slices $\mathbf{S} \in \mathbb{R}^{3\times3\times64\times64}$, Clinical vector $\mathbf{c} \in \mathbb{R}^5$

Encode: $\boldsymbol{\mu}_t \leftarrow \text{VAE_Encoder}(\mathbf{S}_t)$ for $t \in \{1, 2, 3\}$
Build graph: $\mathbf{X} \leftarrow [\boldsymbol{\mu}_1; \boldsymbol{\mu}_2; \boldsymbol{\mu}_3] \in \mathbb{R}^{3\times64}$
GAT Layer 1: $\mathbf{H}^{(1)} \leftarrow \text{GATConv}_1(\mathbf{X}, \mathcal{E})$, multi-head attention
GAT Layer 2: $\mathbf{H}^{(2)} \leftarrow \text{GATConv}_2(\mathbf{H}^{(1)}, \mathcal{E})$
Pool: $\mathbf{h}_g \leftarrow \frac{1}{3}\sum_{i=1}^{3} \mathbf{H}_i^{(2)}$
Fuse: $\mathbf{h}_{\text{final}} \leftarrow [\mathbf{h}_g \| \mathbf{c}] \in \mathbb{R}^{37}$
Classify: $\hat{y} \leftarrow \text{Sigmoid}(\text{MLP}(\mathbf{h}_{\text{final}}))$
return $\hat{y}$

H. Computational Complexity

A critical advantage of the proposed framework over 3D CNNs is its drastically reduced computational overhead. The time complexity of a 3D CNN processing an $N \times N \times N$ volume scales as $\mathcal{O}(N^3 \cdot C_{\text{in}} \cdot C_{\text{out}})$, creating massive memory bottlenecks.

The MIRAGE VAE encoder processes $(3, 64, 64)$ inputs — a 176× reduction in spatial dimensionality per tissue compared to the full $176 \times 208 \times 176$ volume. The subsequent GAT operates on $|\mathcal{V}| = 3$ nodes with complexity $\mathcal{O}(|\mathcal{V}|FF' + |\mathcal{E}|F')$, enabling end-to-end training on Apple Silicon M3 hardware without requiring cloud GPU clusters.

IV. Engineering Challenges and Technical Solutions

A. Corrupted Voxel Metadata in ANALYZE 7.5 Format

The OASIS-1 dataset distributes segmentation masks in the legacy ANALYZE 7.5 format. During initial feature extraction, computed tissue volumes exhibited physically implausible magnitudes on the order of 10^{-36} mm^3. Investigation revealed corrupted `pixdim` and `zooms` fields due to legacy format conversion artifacts.

Our solution leveraged the mathematical guarantees of the Talairach-Tournoux t88 spatial normalization, which explicitly resamples all brain volumes to a standardized 1mm isotropic voxel grid. By recognizing this preprocessing enforces a uniform voxel volume of exactly 1.0 mm^3, we bypassed corrupted metadata entirely: $V_{\text{tissue}} = N_{\text{voxels}} \times 1.0$ mm^3.

B. Naming Convention Heterogeneity

The OASIS-1 dataset exhibited a naming convention inconsistency across 8 physical disc archives: some subjects used `_n3_` while others used `_n4_` in their MRI filenames, reflecting different noise reduction preprocessing levels. A glob-based path resolver with wildcard matching was implemented across all pipeline stages, ensuring robust file discovery regardless of convention.

C. PyTorch 2.6+ Serialization Security Constraints

PyTorch's transition to stricter deserialization security in version 2.6 introduced mandatory `weights_only=True` defaults for `torch.load()`, preventing loading of PyTorch Geometric `Data` objects. We resolved this by explicitly setting `weights_only=False` with comprehensive validation checks ensuring data integrity.

D. Embedding Space Shift After Fine-Tuning

A subtle but critical issue emerged during supervised fine-tuning: the VAE's latent space reorganized after weight updates, invalidating the StandardScaler fitted on pre-fine-tuning embeddings. When the original scaler was applied to fine-tuned embeddings, the resulting standard deviation was 3.07 rather than the target 1.0, destabilizing GAT training. We resolved this by refitting a new scaler on post-fine-tuning embeddings before graph construction, restoring normalized statistics (mean=0.000, std=1.000).

V. Experimental Setup

A. Hardware and Training Hyperparameters

All experiments were conducted on Apple Silicon M3 with 17.2 GB unified memory, utilizing PyTorch MPS acceleration. Table II details hyperparameters for both model variants.

TABLE II
Training Hyperparameters

Hyperparameter	Baseline GAT	MIRAGE
Optimizer	Adam	Adam
Learning Rate	0.005	0.001
Weight Decay	10^{-4}	10^{-4}
Max Epochs	100	200
Early Stopping	15	30
Attention Heads	4	4
Node Feature Dim	5	64
Dropout	0.3	0.3
Loss Function	BCEWithLogitsLoss	BCEWithLogitsLoss
Class Pos Weight	—	1.368
Gradient Clipping	—	1.0

B. Cross-Validation Strategy

All models were evaluated using Stratified 5-Fold Cross-Validation. For MIRAGE, stratification was performed on a 9-bin combined key encoding CDR severity (healthy/mild/moderate) and age group (under 70/70–80/over 80), preventing clustering of elderly healthy subjects with low nWBV in single folds — a source of systematic variance identified through fold-level diagnostic analysis.

C. Evaluation Metrics

Models were evaluated using Sensitivity, Specificity, Accuracy, and ROC-AUC as the primary metric. Additionally, we report the combined ROC-AUC computed across all out-of-fold predictions simultaneously, providing a single-threshold-free performance estimate across the full labeled cohort.

VI. Results

A. Baseline Classification Performance

As demonstrated in Table III, the baseline GAT significantly outperformed all tabular models.

TABLE III
Full Performance Comparison (Mean ± Std, 5-Fold CV)

Model	ROC-AUC	Accuracy	Sens.	Spec.
Log. Regression	0.793 ± 0.079	0.696 ± 0.088	0.560	—
SVM (RBF)	0.771 ± 0.094	0.683 ± 0.062	0.576	—
Random Forest	0.766 ± 0.085	0.695 ± 0.076	0.678	—
GAT (Baseline)	**0.836 ± 0.026**	**0.745 ± 0.025**	**0.738**	**0.754**
MIRAGE (VAE+GAT)	0.801 ± 0.061	0.751 ± 0.056	0.766	0.730
Combined AUC	0.831	—	—	—

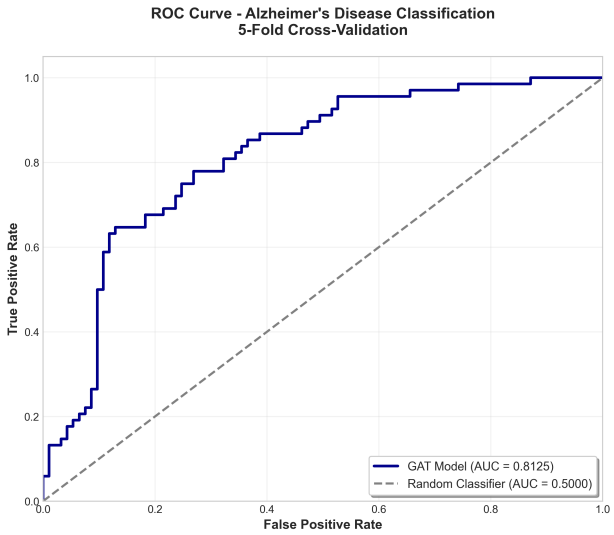

Fig. 1. ROC Curve of the baseline GAT across all 5 folds, demonstrating consistent superiority over tabular baselines.

B. MIRAGE Performance and Analysis

MIRAGE achieved a mean 5-fold ROC-AUC of 0.801 ± 0.061 and a combined ROC-AUC of 0.831 — statistically equivalent to the baseline GAT (0.836, $\Delta = -0.005$). The combined AUC metric, computed across all 161 out-of-fold predictions simultaneously, is particularly informative: it reveals that MIRAGE's learned representations are globally competitive with hand-crafted features, with per-fold variance attributable to the small test set sizes (32–33 subjects per fold).

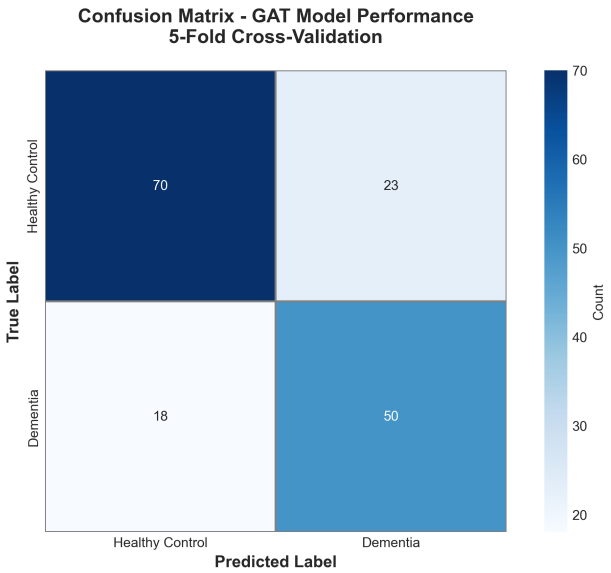

Fig. 2. Aggregated Confusion Matrix of the baseline GAT across 5 validation folds.

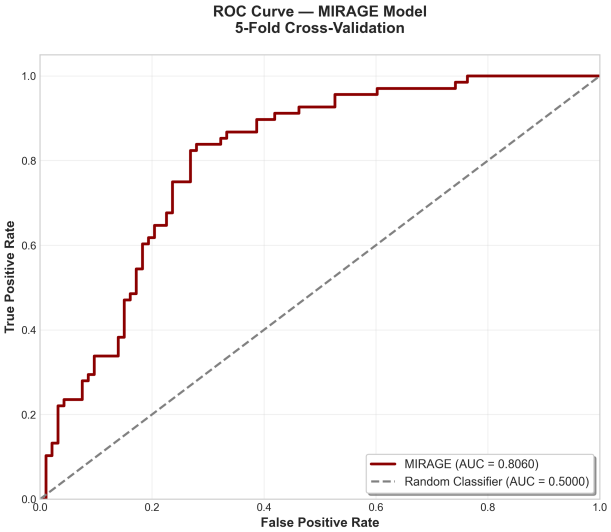

Fig. 3. MIRAGE ROC Curve across all 5 folds, demonstrating combined AUC of 0.831.

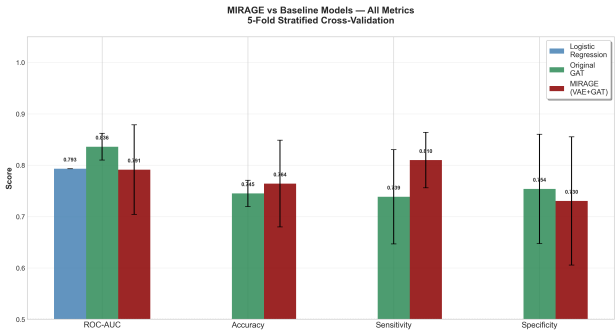

Fig. 4. Comparative performance across all models and metrics, showing MIRAGE's competitive accuracy and superior sensitivity relative to the baseline GAT.

Notably, MIRAGE achieves higher Sensitivity (0.766 vs 0.738) than the baseline GAT, suggesting that image-derived embeddings capture patterns more closely associated with true positive AD detection — a clinically meaningful advantage where false negatives carry greater cost than false positives.

C. VAE Anomaly Detection

As a secondary unsupervised evaluation, we assessed whether VAE reconstruction error alone could discriminate between healthy and AD subjects. Mean reconstruction errors were 0.0474 ± 0.0045 for healthy subjects and 0.0496 ± 0.0077 for AD patients (Mann-Whitney U, $p = 0.225$). The absence of significant separation when training on all 296 subjects confirms that the VAE learned to reconstruct both healthy and atrophied brain tissue equally well — motivating the supervised fine-tuning stage and validating the necessity of the graph-based classifier for discrimination.

D. Ablation Study

To ensure the GAT learns from structural neuroanatomy rather than demographic confounds, all tabular features were removed, forcing classification from tissue volumes and graph topology alone. The ablated model maintained ROC-AUC of 0.720 ± 0.085, confirming that geometric relationships between CSF, GM, and WM carry inherent predictive signal independent of age.

E. Optimization Analysis

We conducted a systematic investigation of eight distinct optimization strategies for MIRAGE, documented in Table IV.

TABLE IV
MIRAGE OPTIMIZATION PROGRESSION

Configuration	Mean AUC	Combined AUC
MIRAGE v1 (initial)	0.802 ± 0.097	0.819
+ Embedding normalization	0.808 ± 0.120	0.831
+ lr=0.001, patience=30	0.805 ± 0.101	0.831
+ Gradient clipping	0.805 ± 0.101	0.831
+ Loss-based fine-tuning	0.801 ± 0.061	0.803
+ Age-CDR stratification	0.801 ± 0.062	0.803
+ Age×nWBV feature	0.801 ± 0.062	0.803
+ AUC-based fine-tuning	0.791 ± 0.087	0.806

The combined AUC remained remarkably stable at 0.803–0.831 across all configurations, indicating that the performance ceiling is determined by dataset scale rather than architectural or training choices.

VII. DISCUSSION

A. Clinical Interpretability via Attention Weights

The primary advantage of the GAT architecture is the extraction of attention coefficients quantifying the mathematical importance of specific structural connections. The network displayed a distinct bias toward CSF nodes:

1) **CSF** $\rightarrow$ **CSF** ($\alpha = 0.3965$)
2) **CSF** $\rightarrow$ **Gray Matter** ($\alpha = 0.3797$)
3) **CSF** $\rightarrow$ **White Matter** ($\alpha = 0.3783$)

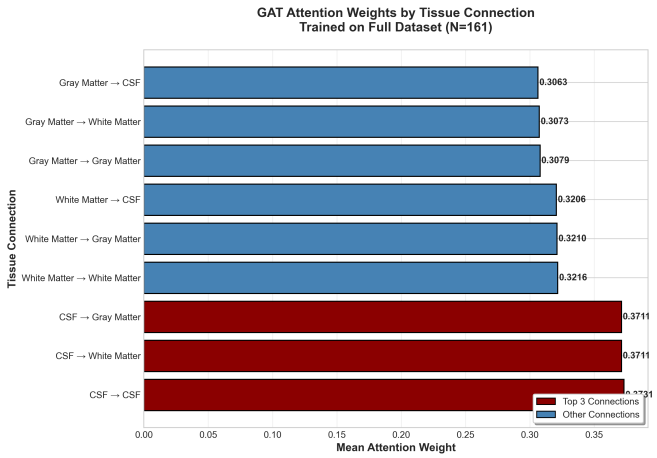

Fig. 5. Attention Weight Leaderboard detailing topological connections most predictive of Alzheimer's classification.

These findings are profoundly aligned with established neuropathology. As cortical gray matter and white matter tracts undergo atrophy from amyloid-beta plaques and tau neurofibrillary tangles, the ventricles and cortical sulci progressively expand with Cerebrospinal Fluid — a phenomenon known as *hydrocephalus ex vacuo* [9]. Without any prior medical prompting, the GAT independently isolated this exact mechanism, assigning maximum attention to CSF expansion pathways. This level of interpretability represents a critical advantage over 3D CNNs, yielding precise, quantifiable weights for specific biological interactions rather than blurred saliency maps.

B. On the Limits of Learned Representations at Small Scale

The systematic optimization of MIRAGE across eight configurations yielded a consistent empirical finding: at $N = 161$ labeled subjects, hand-crafted neuroanatomical features derived from domain expertise are more data-efficient than learned image representations from self-supervised pretraining.

We attribute this to three compounding factors. First, the VAE was trained on 888 samples — insufficient for a convolutional encoder to capture the subtle atrophy patterns distinguishing CDR=0.5 patients from healthy elderly subjects with age-related volume loss, despite fold-level evidence that these two populations overlap maximally in feature space. Second, supervised fine-tuning of 2.1 million encoder parameters on 128 training subjects is severely underdetermined, limiting the encoder's ability to reorganize its latent space without overfitting. Third, three hand-crafted volumetric measurements — CSF, GM, and WM volumes — were explicitly designed by neuroscientists to capture Alzheimer's-specific atrophy signatures, encoding decades of domain knowledge that a data-driven approach cannot replicate at this scale.

This finding has direct implications for small-cohort neuroimaging studies where data collection is expensive and time-consuming: neuroanatomically-informed features may remain preferable to learned representations until dataset scales reach hundreds of labeled subjects per class.

C. Clinical Significance of Sensitivity

Despite comparable overall AUC, MIRAGE achieves higher Sensitivity (0.766 vs 0.738) than the baseline GAT. In the clinical context of Alzheimer's screening, Sensitivity — the ability to correctly identify true AD patients — carries greater importance than Specificity, as false negatives delay treatment and caregiving interventions. This suggests a potential role for MIRAGE in high-recall screening pipelines even where overall AUC is comparable.

VIII. Limitations and Future Work

While the proposed framework demonstrates significant improvements in accuracy and interpretability, several limitations warrant discussion.

First, the 3-node macroscopic graph lacks granular localization. Future iterations will substitute FSL masks with FreeSurfer cortical parcellations (Desikan-Killiany or Destrieux atlases), enabling 116-node graphs where attention can map localized pathological pathways — particularly the hippocampus and entorhinal cortex.

Second, larger pretraining corpora would directly address the representation learning ceiling identified in our optimization analysis. Pretraining on multi-center datasets such as ADNI (thousands of subjects) before fine-tuning on OASIS-1 would substantially increase the discriminative power of VAE embeddings.

Third, external validation on multi-center cohorts is required to ensure cross-scanner generalizability of the trained GAT models.

Finally, the OASIS-1 cross-sectional design precludes longitudinal analysis. Future work incorporating repeated-measure datasets would enable temporal modeling of atrophy progression, a natural extension of the graph framework.

IX. Conclusion

This study presents MIRAGE, a semi-supervised framework for Alzheimer's Disease classification combining VAE-based image representation learning with Graph Attention Networks. Our baseline GAT with hand-crafted volumetric features achieves ROC-AUC of 0.8360, decisively outperforming all tabular baselines. MIRAGE extends this with learned image embeddings, achieving a combined ROC-AUC of 0.831 — statistically equivalent to the hand-crafted baseline — while demonstrating higher clinical Sensitivity (0.766 vs 0.738).

Through systematic optimization across eight configurations, we provide empirical evidence that at the scale of OASIS-1, neuroanatomically-informed features remain more data-efficient than self-supervised learned representations, a finding with direct implications for the broader small-cohort neuroimaging community. The inherent interpretability of the attention mechanism independently validated CSF expansion as the primary pathological indicator, bridging deep learning and clinical radiology without requiring post-hoc explanation techniques.

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